Fourth Degree Function Report

Polynomial

$$f(q) = aq^4 + bq^3 + cq^2 + dq + e$$

Naive

$$g_{\text{naive}}(X) = X^4 a + X^3 b + X^2 c + Xd + e$$

$$\mathbb{E}[g_{\text{naive}}(X)] = 24a{\frac{\Delta}{\epsilon}}^4 + aq^4 + bq^3 + 2{\frac{\Delta}{\epsilon}}^2 \left(6aq^2 + 3bq + c\right) + cq^2 + dq + e$$

$$\begin{aligned}\operatorname{Var}(g_{\text{naive}}(X)) = 2{\frac{\Delta}{\epsilon}}^2 \Bigg(&19872a^2{\frac{\Delta}{\epsilon}}^6 + 16a^2q^6 + 24abq^5 + 16acq^4 + 8adq^3 + 9b^2q^4 \\ &+ 12bcq^3 + 6bdq^2 + 24{\frac{\Delta}{\epsilon}}^4 \left(408a^2q^2 + 204abq + 28ac + 15b^2\right) \\ &+ 2{\frac{\Delta}{\epsilon}}^2 \left(372a^2q^4 + 372abq^3 + 156acq^2 + 48adq + 81b^2q^2 \right. \\ &\left. + 54bcq + 12bd + 5c^2\right) + 4c^2q^2 + 4cdq + d^2\Bigg)\end{aligned}$$

$$\begin{aligned}\operatorname{MSE}(g_{\text{naive}}) = 2{\frac{\Delta}{\epsilon}}^2 \Bigg(&20160a^2{\frac{\Delta}{\epsilon}}^6 + 16a^2q^6 + 24abq^5 + 16acq^4 + 8adq^3 + 9b^2q^4 \\ &+ 12bcq^3 + 6bdq^2 + 360{\frac{\Delta}{\epsilon}}^4 \left(28a^2q^2 + 14abq + 2ac + b^2\right) \\ &+ 12{\frac{\Delta}{\epsilon}}^2 \left(68a^2q^4 + 68abq^3 + 28acq^2 + 8adq + 15b^2q^2 + 10bcq \right. \\ &\left. + 2bd + c^2\right) + 4c^2q^2 + 4cdq + d^2\Bigg)\end{aligned}$$

$$\operatorname{Bias}(g_{\text{naive}}) = 2{\frac{\Delta}{\epsilon}}^2 \left(12a{\frac{\Delta}{\epsilon}}^2 + 6aq^2 + 3bq + c\right)$$

Unbiased

$$g_{\text{unb}}(X) = X^4 a + X^3 b + X^2 c + Xd - 2{\frac{\Delta}{\epsilon}}^2 \left(6X^2 a + 3Xb + c\right) + e$$

$$\mathbb{E}[g_{\text{unb}}(X)] = aq^4 + bq^3 + cq^2 + dq + e$$

$$\mathrm{Var}(g_{\mathrm{unb}}(X)) = 2\frac{\Delta}{\epsilon}^2 \Bigg(13248a^2\frac{\Delta}{\epsilon}^6 + 16a^2q^6 + 24abq^5 + 16acq^4 + 8adq^3 + 9b^2q^4 + 12bcq^3 + 6bdq^2 + 36\frac{\Delta}{\epsilon}^4 \left(184a^2q^2 + 92abq + 12ac + 7b^2\right) + 2\frac{\Delta}{\epsilon}^2 \left(276a^2q^4 + 276abq^3 + 108acq^2 + 24adq + 63b^2q^2 + 42bcq + 6bd + 5c^2\right) + 4c^2q^2 + 4cdq + d^2\Bigg)$$

$$\mathrm{MSE}(g_{\mathrm{unb}}) = 2\frac{\Delta}{\epsilon}^2 \Bigg(13248a^2\frac{\Delta}{\epsilon}^6 + 16a^2q^6 + 24abq^5 + 16acq^4 + 8adq^3 + 9b^2q^4 + 12bcq^3 + 6bdq^2 + 36\frac{\Delta}{\epsilon}^4 \left(184a^2q^2 + 92abq + 12ac + 7b^2\right) + 2\frac{\Delta}{\epsilon}^2 \left(276a^2q^4 + 276abq^3 + 108acq^2 + 24adq + 63b^2q^2 + 42bcq + 6bd + 5c^2\right) + 4c^2q^2 + 4cdq + d^2\Bigg)$$

Comparison

$$\text{mean gap} = 2\frac{\Delta}{\epsilon}^2 \left(-12a\frac{\Delta}{\epsilon}^2 - 6aq^2 - 3bq - c\right)$$

$$\text{variance gap} = 24\frac{\Delta}{\epsilon}^4 \Bigg(-552a^2\frac{\Delta}{\epsilon}^4 - 16a^2q^4 - 16abq^3 - 8acq^2 - 4adq - 3b^2q^2 - 2bcq - bd - \frac{\Delta}{\epsilon}^2 \left(264a^2q^2 + 132abq + 20ac + 9b^2\right)\Bigg)$$

$$\text{MSE gap} = 4\frac{\Delta}{\epsilon}^4 \Bigg(-3456a^2\frac{\Delta}{\epsilon}^4 - 132a^2q^4 - 132abq^3 - 60acq^2 - 24adq - 27b^2q^2 - 18bcq - 6bd - 18\frac{\Delta}{\epsilon}^2 \left(96a^2q^2 + 48abq + 8ac + 3b^2\right) - c^2\Bigg)$$

$$\mathrm{Bias}(g_{\mathrm{naive}})^2 = 4\frac{\Delta}{\epsilon}^4 \Bigg(144a^2\frac{\Delta}{\epsilon}^4 + 36a^2q^4 + 36abq^3 + 24a\frac{\Delta}{\epsilon}^2 \left(6aq^2 + 3bq + c\right) + 12acq^2 + 9b^2q^2 + 6bcq + c^2\Bigg)$$